

Deep Learning Classification of Subtypes of Luminal Breast Cancer from Immunohistochemistry Slides

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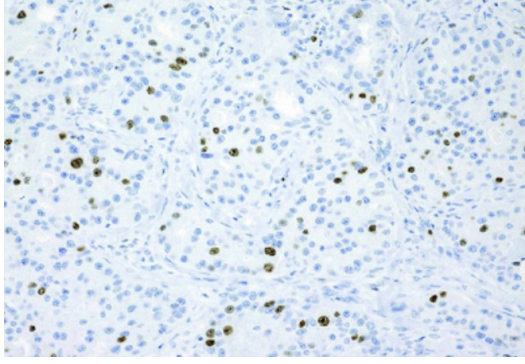
Abstract. This paper proposes deep-learning models to assist in classifying subtypes of luminal breast cancer from immunohistochemistry slides. The classification has traditionally been done by pathologists who evaluate Ki67 indexes by manually counting positive and negative cells that are dyed in different colours in the immunohistochemistry slides. This manual process is error-prone and time-consuming. Therefore, this paper explores the possibility of automatically classifying immunohistochemistry slides. The performances of 4 pre-trained, deep convolutional networks are compared. Based on the good performance of the models, we conclude that the task of Ki67 classification for determining subtypes of luminal breast cancer is well-suited for deep learning models and that for this reason, manual counting of Ki67-positive cells are neither needed nor desirable.

Keywords: Luminal breast cancer, Ki67 index, Image classification, Convolutional neural networks

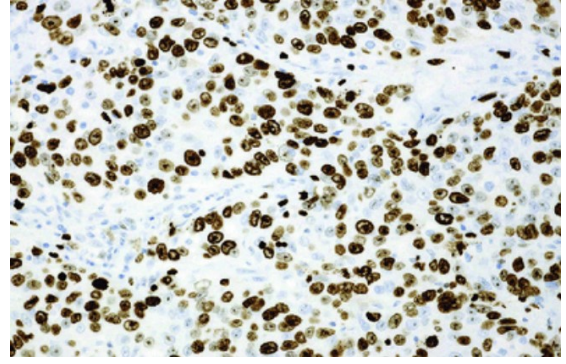
1 Introduction

Ki67 is widely used as a proliferation marker in breast cancers [11]. A higher Ki67 index indicates a higher tumour cell proliferation rate, which in turn indicates a more aggressive tumour type [5]. The proliferation rate is defined the percentage of positively stained Ki67 cells among the total number of malignant cells. Usually, pathologists evaluate the Ki67 index from an immunohistochemistry (IHC) slide. An IHC slide is a type of histopathological slide where immunostaining is applied. Figure 1 shows two images of IHC slides with high and low Ki67 indexes.

To prepare an IHC slide for studying breast cancer, the first step is to take a tissue sample from the tumour specimen, typically obtained through biopsies or surgical resections. This sample is sent to a pathology lab, where the sample is embedded in paraffin after formalin fixation. The resulting paraffin block is then cut into fine slices that are mounted in glass. These glass-mounted slides are called histopathological slides. To bring out the properties of the cells, the tissue needs to be dyed with stains that highlight the structures of interest. The two most common staining protocols are hematoxylin and eosin (H&E) and IHC straining. In H&E straining, hematoxylin colours the nuclei blue and purple by binding to the DNA, and the eosin colours cytoplasm and some other structures pink by binding to certain proteins. On the other hand, in IHC straining, specific proteins can be detected by using dyes that bind to specific primary antibodies. Veta et al. [18] give an overview of the issues in tissue preparation and staining for breast cancer histopathology slides.



(a) Ki67 - Low



(b) Ki67 - High

Fig. 1: Zoomed views on two IHC stained slides showing different intensities of Ki6, where brown colour indicates positively stained Ki67 cells and blue colour indicates negatively stained cells. Fig(1a) Low Ki67 value (less than 10%). Fig(1b) High Ki67 value (almost 80%).

The evaluation of Ki67 starts with an IHC slide where negative cells are dyed in blue and positive cells are dyed in brown shades. Several methods have been proposed to determine the Ki67 value. Among these, the three most common approaches are following [6].

1. In the first approach, one estimates the percentage of positive Ki67 cells among a fixed number of tumour cells within a hotspot region. Such a hotspot region is an area in the slide with a high density of Ki67-positive tumour cells relative to adjacent areas. This is done by counting the number of positive cells among a fixed number of cells (for example, a thousand cells). The selection of the hotspot area and the predefined number of cells may considerably impact the estimated Ki67 value.
2. In the second approach, multiple hotspots are selected, and each hotspot's average percentage of positive cells is estimated. The overall estimate of the Ki67 index is then obtained by averaging the percentages from each of the selected hotspot areas. Unlike in the first approach, the average percentage of positive cells is estimated within all tumour cells within the hotspot. However, the selection of hotspot regions is still subjective, and this selection might still lead to different estimates on the same slide by different pathologists.
3. In the third approach, the fraction of positive cells within the whole slide is estimated. This includes the whole slide, unlike in the previous two approaches. While this removes the subjective selection of hotspot areas, it introduces a new problem: manually counting all cells is time-consuming and error-prone.

Lashen et al. [6] recommend using the first method as the most feasible approach. However, in practice, none of these approaches might be strictly followed, and the Ki67 index is often estimated using various techniques such as “eye-balling estimation” and “vision counting”. In summary, all the methods described above are tedious and involve human judgment, which makes them error-prone and subjective with limited reproducibility [6].

As mentioned before, Ki67 is of great clinical relevance. Among other uses [9, 10], Ki67 assessment is one of the criteria to distinguish between Luminal A and Luminal B cancers [17], where Luminal B is a high-risk type of cancer, and those patients may need additional treatment like chemotherapy [12]. Several cutoff points for distinguishing between the luminal subtypes are in common use; in the following, we will use 20 per cent [7]. The clinical guidelines for early-stage breast cancer published by the European Society for Medical Oncology (ESMO) in 2019 also proposes a cut-off of 20%. Since the manual assessment of Ki67 from IHC slides has the issue of reproducibility between different observers and is a time-consuming process, this paper aims to develop a method to automatically process and analyze IHC slides for the classification of the genetic breast cancer subtypes Luminal A and Luminal B. One approach to solve this problem would be to start by automatically counting the Ki67-positive cells, either by thresholding (implemented in software like ImageJ [1] and Aperio Image Scop [14]) or using machine learning methods [21], and in a second step compare the count with the threshold value. Reliable, automated counting of Ki67-positive cells is an ongoing research area, and none of the solutions are completely satisfactory [3]. In this paper, we take a different

approach, where instead of segmenting and counting cells first and classifying later, we use deep learning to classify images into a high Ki67 (positive) and low Ki67 (negative) group directly.

The outline of the paper is as follows. Section 2 describes the background of the data set, the data preprocessing, the validation strategy, and the deep learning models that are used for the classification. Section 3 presents the results, and Section 4 ends with a discussion and gives conclusions.

2 Methodology

2.1 Data

A total of 129 histopathologic slides from breast cancer patients were prepared for evaluation, with 80 positive cases and 49 negative controls. The resolution of the images is 300 pixels per inch. Data were collected at the Amrita Institute of Medical Sciences (AIMS), a super-specialty tertiary-care hospital in South India. The study was reviewed and approved by the Institutional Review Board of AIMS.

Two expert pathologists assessed the percentage of invasive cancer cells expressing Ki67 in each slide. In more detail, the pathologists independently scored 500 tumour cells in three to five hot spots at 400x magnification. Nuclear staining of any intensity was considered positive. If the two pathologists came to different assessments on whether Ki67 exceeded 20 percent in a slide, the judgment was made by a third pathologist who independently made an additional assessment. The average Ki67 value of the two pathologists was recorded, and the classification of whether or not Ki67 exceeded 20 per cent was taken for the analysis.

2.2 Preprocessing

During the preprocessing phase, various transformation techniques were used to prepare the data for deep learning using the PyTorch library. From the available dataset, new images were generated using data augmentation techniques to increase the size of the training set (see description of the validation strategy in the next section). In total, 400 images were generated. For each image in the training the below augmentation methods were used to create new images which had different orientations, contrast levels, to create a challenging dataset which can be used to train the models efficiently.

Resize() This transformation is commonly used in image preprocessing and computer vision tasks. It is used to resize input images to a specified target size. It ensures that all the input images are of the same size so that the deep learning models can work on a uniform input size.

RandomRotation() The transformation rotates input images. The degree of rotation is chosen randomly from a specified range. This introduces diversity into the training data, ensuring that the model does not depend on the specific orientation of the slides.

ColorJitter() This transformation applies random variations to an image's brightness, contrast, saturation, and hue within specified limits. By doing so, the model can adapt to different lighting issues that may be caused when a raw input is given.

RandomHorizontalFlip() This transformation randomly flips the images horizontally. It horizontally mirrors the image, thereby shifting the orientation of the images. Including augmented images that are randomly flipped into the training data ensures that the model is robust with respect to unfamiliar orientations.

RandomVerticalFlip() Similar to RandomHorizontalFlip, this transformation randomly flips the input images vertically. It vertically mirrors the image, thereby again changing the orientation of the images.

ToTensor() The deep learning models expect their input as tensor objects. ToTensor transforms an image to such a PyTorch tensor. The tensor can then be fed into the model for training.

2.3 Models

Deep learning is a sub-field of artificial intelligence and more specifically, machine learning. Artificial neural networks are the most prominent class of deep learning models. Such networks are computational models inspired by a human brain's functions and structure. Deep learning models have been successfully applied across various domains, from computer vision and natural language processing [16] to intrusion detection [20], and healthcare [19]. This section will briefly summarise the concepts of deep learning, neural networks, and convolutional neural networks for image processing.

A neural network consists of different interconnected nodes, which are also known as artificial neurons. These neurons are organized in layers, typically consisting of an input layer, one or more hidden layers, and one output layer. The strength of each connection between two neurons is determined by the weight associated with the connection. Neural networks are highly versatile because of their capability to learn complex patterns and representations from data. The process of training a neural network involves adjusting the weights of its connections to minimize the difference between the predicted output and the observed outcomes. Different iterative algorithms are used for this optimisation, such as backpropagation and gradient descent. In the context of fitting parameters of neural networks, each iteration is called an epoch.

Convolutional Neural Networks (CNNs) are specially designed deep neural networks which are used to process grid-like data, such as images and video frames. CNNs have revolutionized computer vision tasks by incorporating convolutional layers that apply learnable filters to the input data. These filters capture the local patterns and features, and then the filters build a hierarchical representation of the input. The ability of CNNs to preserve the spatial relationships between pixels makes them well-suited for image-related tasks. Such networks consist of convolutional layers, pooling layers, and the fully-connected layers for making predictions. Due to this architecture enables CNNs to automatically detect the edges, textures, and higher-level features like object parts and shapes, which is essential for image classification and segmentation. CNNs have been used in a wide range of applications, including medical image analysis, remote sensing, autonomous driving, and many others.

Instead of training CNNs from scratch on each new dataset, it has now become common to use pre-trained models that determine the network architecture and contain pre-trained neuron connections. These pre-trained models have been created on large datasets, and transferring them to the specific application domain generally improves the performance for tasks such as image classification and segmentation.

We have used four suitable pre-trained models for the classification of breast cancer IHC slides. The pre-trained models are ResNet-50, DenseNet-121, MobileNet V2, and Inception V3. We will now briefly discuss the capability and performance of these four models.

ResNet-50 [2] The Deep Residual Network 50 (ResNet-50) is one of the variants of the ResNet family. ResNet-50 has a 50-layer convolutional neural network in which 48 are convolutional layers, then one MaxPool layer, and one average pool layer. The architecture features residual blocks with identity shortcuts. The convolutional layers of ResNet-50 have already learned a wide range of low to high-level features from the massive datasets available, such as ImageNet. ResNet-50 main advantage is its capability to serve as a powerful feature extractor for a broad range of variety of computer vision tasks. The residual blocks are the blocks that allow for shortcuts, i.e., skip connections, which make it easier for the gradients to flow and allow networks to learn complex transformations and stabilise the learning process, as it also enhances the ability of the blocks to learn identity mapping. The deep architecture of ResNet-50 makes it well-suited for feature extraction in complicated images, such as breast cancer IHC slides.

DenseNet-121 [4] The Densely Connected Convolutional Network: The architecture of DenseNet-121 is known for its densely connected blocks. It contains a total of 121 layers in which 117 convolutions, 3 transitions, and 1 classification layers are present. In the traditional convolutional neural networks, the layers are connected sequentially. Whereas in the DenseNet model, each layer is connected to every other layer. As the name suggests, this dense connectivity of the layers promotes feature reuse and facilitates gradient flow during training, resulting in a compact and highly efficient network. Transition blocks help reducing the the spatial dimension, which refers to the height and width of the activation maps. It reduces this spatial dimension using max pooling or average pooling and convolutions. Reduction in spatial dimension reduces

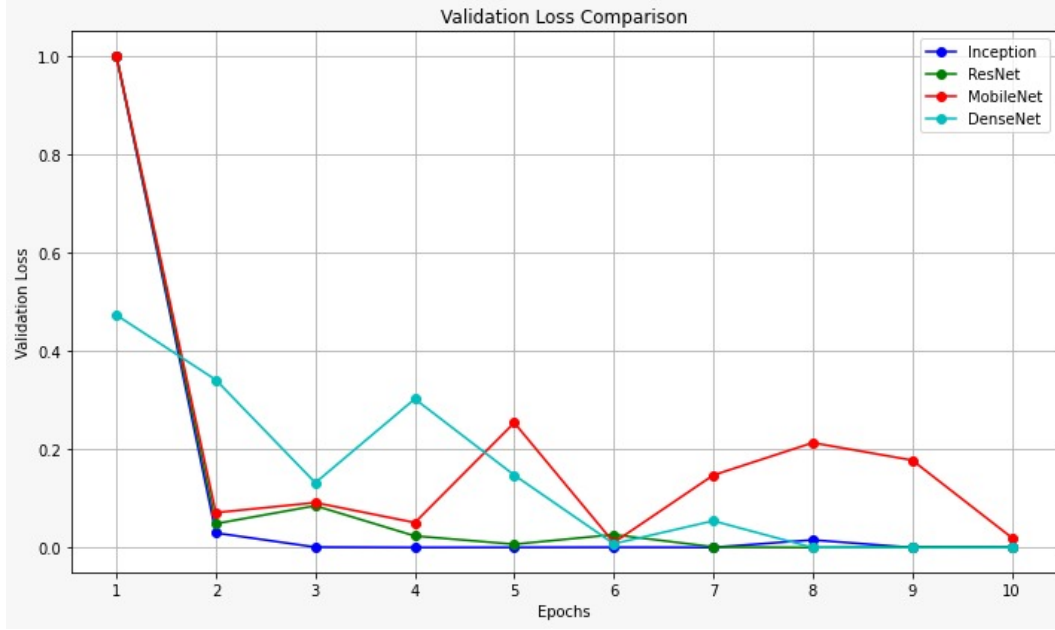


Fig. 2: Validation loss comparison for different models

the computational cost and makes sure that the model does not overfit. While pre-training the DenseNet-121 models, they capture a very rich hierarchy of features. Their performance in tasks requiring precise feature discrimination, such as medical image analysis and other fields, is impressive. Dense connectivity of the layer allows for better gradient propagation through the network; this helps in the mitigation of the vanishing gradient problem, making DenseNet-121 a valuable pre-trained model in deep learning. The densely connected blocks of the model enable the efficient extraction of fine-grained features. In Ki-67 breast cancer cell recognition also, this model works with good accuracy at capturing variations in cell morphology and staining patterns.

MobileNet V2 [13] MobileNet V2 is a popular deep-learning model used in the field of computer vision; as the name suggests, it is primarily used for mobile devices and edge devices. MobileNet-v2 is a convolutional neural network that is 53 layers deep. It has depth-wise separable convolutions, which reduce the computational cost at the same time maintaining high accuracy. Depth-wise convolution is used in lightweight models, which splits the convolution process into 2 steps. First is depth-wise convolution, in which each input channel is convolved with its own set of filters and point-wise convolution which combines the output channel using linear combinations. This can reduce the computational cost and act as regularization, which makes it harder for the model to overfit. This makes the model lightweight and easy to train and adjust to new data as compared to other models available for the same set of tasks.

Inception V3 [15] It is a CNN model created by Google for computer vision tasks. It is 48 layers deep. It uses inception modules that incorporate multiple filter sizes that run in parallel to capture the different features. Compared to GoogleNet, another CNN model developed at Google, Inception incorporates label smoothing and factorized convolutional. Label smoothing is a regularization technique that generalizes the one-hot labels by redistributing some probability mass. This ensures that the model is not overconfident in its prediction and reduces the chance of the model overfitting. Inception V3 is widely used in the medical field, where its segmentation capability captures fine-grained features.

2.4 Validation scheme

Initially, the data were divided into training and test data, with 70% for training and the remaining 30% for testing. After preprocessing and augmentation, 20% of the augmented training set was set aside as a

validation set. During each epoch of the iterative training of the models, the validation set was used to validate the model's current performance. Changes in the validation loss from one epoch to the next were used as stopping criteria. Based on the validation loss, we also adjusted the learning rate to optimize the model's training speed. This iterative validation and adjustment cycle ensured that the model constantly improved its performance.

After training the models, the following metrics were used to evaluate the models' performances on the test data:

Accuracy This measures the overall accuracy of a model's predictions. Put more precisely, the measure gives the percentage of correct classifications. It is calculated as follows:

$$\text{Accuracy} = \frac{\#\{\text{correct predictions}\}}{\#\{\text{all predictions}\}}$$

Precision This is another important indicator of a machine learning model's performance. It measures the percentage of correct positive model predictions. It is calculated as follows:

$$\text{Precision} = \frac{\#\{\text{true positive predictions}\}}{\#\{\text{true positive predictions}\} + \#\{\text{false positive predictions}\}}$$

Recall Recall is also called sensitivity. It measures the proportion of the true predictions among positive cases and thereby provides a measure of the model's sensitivity in detecting positive cases:

$$\text{Recall} = \frac{\#\{\text{true positive predictions}\}}{\#\{\text{positive cases}\}}$$

F1-Score The F1-Score is the harmonic mean of precision and recall. It balances precision and recall, making it a valuable metric for assessing model performance in imbalanced data sets. It helps to analyse false positives and false negatives. It is calculated as follows:

$$\text{F1-Score} = \frac{(\text{Precision} \times \text{Recall})}{(\text{Precision} + \text{Recall})}$$

2.5 Implementation

We implemented the models in Python due to its widespread popularity and the extensive ecosystem of libraries and frameworks, simplifying the complexities associated with deep learning tasks. More specifically, we used the PyTorch and Torchvision for working with the deep learning models. PyTorch is one of the most efficient deep learning frameworks for Python, and it seamlessly integrates with GPUs. Torchvision is a popular package in the PyTorch framework, which has a wide range of tools and utilities for working with computer-vision tasks. It contains pre-trained models we have used in this project, transformation techniques used to create the augmented images, and a data loader, which makes it easy to batch-load the data for training, validation and testing the model. Hyperparameter tuning is the process of finding the optimal hyperparameter. We adjusted hyperparameters during the training phase by monitoring the validation loss on the validation set.

3 Results

Figure 2 shows the validation loss of the different models during the model training. We see that each model converged very quickly and that all models' validation loss goes to zero. This means that, at least on the training data, the models can make perfect predictions. The stopping criteria for the models were if the validation loss was not improving with each epoch when compared to the past epoch, the model training would stop because overtraining the model would cause it to overfit and make the accuracy way lower than expected. However, to assess the models' true performance on new data, we compared the models on the test data; the results are shown in Table 1, which lists the model's results on the test set and displays all the key metrics, including accuracy, precision, recall, and F1-Score, for each of the luminal cancer subtype classification models. We can observe here that all 4 models have excellent performance, and we conclude that the task of classifying luminal cancer subtypes can be done efficiently and reliably by deep learning models.

Table 1: Percentages of performance comparison of deep learning models on various metrics for the test data.

	Accuracy	Precision	Recall	F1-Score
ResNet	90.9	90.9	90.9	90.9
Inception	89.1	89.1	89.1	89.1
MobileNet	100.0	100.0	100.0	100.0
DenseNet	92.7	92.7	92.7	92.7

4 Discussion

The traditional method used to evaluate the percentage value of Ki67 is time-consuming and subjective. To manually find the number of positively stained cells from the stained slide requires a lot of concentration and takes a long time. The problem of automatic classification of IHC slides is of considerable relevance. Multiple approaches have been proposed to determine the Ki67 value from IHC slides [18]. Traditional image-segmentation techniques cannot adequately count overlapping cells in slides [21]. Nielson et al. details the recommendations from International Ki67 in Breast Cancer Working Group (IKWG) [11]

Instead of direct assessment of Ki67, some papers consider the classification of images based on features that would traditionally have been derived from a manual assessment of Ki67, for example, the diagnosis of breast cancer in the early stages [22].

This paper considers another such problem that currently requires the manual assessment of Ki67. Instead of segmenting cells and estimating Ki67, the problem is formulated as a binary classification problem. The results show that deep learning models can solve this problem with high accuracy.

In more detail, we find that the specific network architecture is not of paramount importance because all the pre-trained networks can achieve very high accuracy within a relatively short training period. Regarding the feasibility of this approach, we note that the trained models can be applied with little computational demand and a relatively small memory footprint.

However, even with robust feature extraction and classification mechanisms, the challenges of integrating such an automated image classification approach in laboratory practice are multifaceted. Automated histopathologic analysis likely involves laboratory-specific pre-processing of images, and integrating deep learning methods might, therefore, require adaptations to the workflows in pathology laboratories. Furthermore, the lack of standardized Ki67 assessment protocols poses a challenge to the training and validating of the methods in a new laboratory.

Future work might elicit domain-specific and laboratory-specific demands and extend the deep learning models accordingly. Furthermore, future work should explore to what extent the method can be generalized to other laboratories that might follow different guidelines for analyzing IHC slides.

In conclusion, the relevance of the proposed approach lies in its potential to automate routine tasks in breast cancer pathology. Accurate classification of IHC slides is required to diagnose and treat breast cancer patients. Standardizing the preanalytical and analytical variabilities in IHC Ki67 evaluation would minimize the inconsistencies in Ki67 estimation [8]. In this way, this paper provides another small step to bridge the gap between traditional pathology and cutting-edge machine learning, paving the way for efficient, reproducible, and objective diagnoses.

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